

1. $\cos kx$ 求和问题, 晚上讲的公式有误, 现修正一下

已知欧拉公式:

$$e^{ix} = \cos x + i \sin x$$

由此可知:

$$e^{inx} = \cos nx + i \sin nx, e^{-inx} = \cos nx - i \sin nx$$

由此可知,

$$\cos nx = \frac{1}{2}(e^{inx} + e^{-inx}), \sin nx = \frac{1}{2i}(e^{inx} - e^{-inx}),$$

因此

$$\sum_{n=0}^k \cos nx = \sum_{n=0}^k \frac{1}{2}(e^{inx} + e^{-inx})$$

由等差数列求和公式可知,

$$\sum_{n=0}^k \cos nx = \frac{1}{2} \left(\frac{1 - e^{i(k+1)x}}{1 - e^{ix}} \right) + \frac{1}{2} \left(\frac{1 - e^{-i(k+1)x}}{1 - e^{-ix}} \right)$$

再由于

$$\frac{1 - e^{i(k+1)x}}{1 - e^{ix}} = \frac{1 - e^{i(k+1)x}}{e^{i\frac{x}{2}}(e^{-i\frac{x}{2}} - e^{i\frac{x}{2}})} = \frac{e^{-i\frac{x}{2}} - e^{i(k+\frac{1}{2})x}}{-2i \sin \frac{x}{2}}$$

同理,

$$\frac{1 - e^{-i(k+1)x}}{1 - e^{-ix}} = \frac{1 - e^{-i(k+1)x}}{e^{-i\frac{x}{2}}(e^{i\frac{x}{2}} - e^{-i\frac{x}{2}})} = \frac{e^{i\frac{x}{2}} - e^{-i(k+\frac{1}{2})x}}{2i \sin \frac{x}{2}}$$

相加可知,

$$\begin{aligned} \sum_{n=0}^k \cos nx &= \frac{1}{2} \left(\frac{e^{i(k+\frac{1}{2})x} - e^{-i\frac{x}{2}}}{2i \sin \frac{x}{2}} + \frac{e^{i\frac{x}{2}} - e^{-i(k+\frac{1}{2})x}}{2i \sin \frac{x}{2}} \right) \\ &= \frac{1}{2} \left(\frac{2i \sin \frac{x}{2} + 2i \sin (k + \frac{1}{2})x}{2i \sin \frac{x}{2}} \right) = \frac{1}{2} \left(1 + \frac{\sin (k + \frac{1}{2})x}{\sin \frac{x}{2}} \right) \end{aligned}$$

由此可知:

$$\frac{\sin (k + \frac{1}{2})x}{\sin \frac{x}{2}} = 1 + 2 \left(\sum_{n=1}^k \cos nx \right)$$

2. 积分练习题 计算积分 $\int_0^{\frac{\pi}{2}} \cos(2nx) \ln(\cos x) dx$.

解 首先用分部积分法消去 “ln”

$$\begin{aligned} \text{原式} &= \frac{1}{2n} \int_0^{\frac{\pi}{2}} \ln(\cos x) d \sin(2nx) \\ &= \frac{1}{2n} \sin(2nx) \ln(\cos x) \Big|_0^{\frac{\pi}{2}} - \frac{1}{2n} \int_0^{\frac{\pi}{2}} \frac{\sin(2nx)(-\sin x)}{\cos x} dx \\ &= \frac{1}{2n} \int_0^{\frac{\pi}{2}} \frac{\sin(2nx) \sin x}{\cos x} dx. \end{aligned} \tag{1}$$

因为

$$\begin{aligned}\sin(2nx) \cdot \sin x &= \cos(2nx) \cos x - \cos(2n+1)x \\ \frac{1}{2n} \int_0^{\frac{\pi}{2}} \frac{\sin(2nx) \sin x}{\cos x} dx &= \frac{1}{2n} \int_0^{\frac{\pi}{2}} \cos(2nx) dx - \frac{1}{2n} \int_0^{\frac{\pi}{2}} \frac{\cos(2n+1)x}{\cos x} dx,\end{aligned}$$

第一个积分为 0, 第二个积分中令 $x = t - \frac{\pi}{2}$, 可知:

$$\begin{aligned}\frac{1}{2n} \int_0^{\frac{\pi}{2}} \frac{\sin(2nx) \sin x}{\cos x} dx &= \frac{(-1)^{n-1}}{2n} \int_{\frac{\pi}{2}}^{\pi} \frac{\sin(2n+1)t}{\sin t} dt \quad (\text{利用 1 中的求和公式}) \\ &= \frac{(-1)^{n-1}}{2n} \int_{\frac{\pi}{2}}^{\pi} \left[1 + 2 \sum_{k=1}^n \cos(2kt) \right] dt = (-1)^{n-1} \frac{\pi}{4n}.\end{aligned}$$

3. 积分证明题

$f(x)$ 在 $[0, 2]$ 上可积, 且 $\int_0^2 f(x) dx = 0$. 求证: 存在 $a \in [0, 1]$, 使 $\int_a^{a+1} f(x) dx = 0$.

解: 由 $f(x)$ 可积我们知道, $F(x) = \int_x^{x+1} f(t) dt$ 为 $[0, 1]$ 上的连续函数

另一方面, $F(0) + F(1) = \int_0^2 f(x) dx = 0$, 表明 $F(0)$ 与 $F(1)$ 异号, 或同时为零.

利用连续函数的介值性, $F(x)$ 在 $[0, 1]$ 里有零点, 即 $\exists \alpha \in [0, 1]$, 使得 $F(\alpha) = \int_{\alpha}^{\alpha+1} f(x) dx = 0$.